

Ashim Mahajan

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PROFILE

Machine Learning Engineer with experience building end-to-end ML pipelines and deploying production inference systems using FastAPI and Docker. Skilled in OCR/document intelligence, computer vision with transfer learning, and LLM-based text processing. Experienced with AWS services (S3, Lambda, EC2) for scalable ML deployment.

SKILLS

Machine Learning

TensorFlow | PyTorch |
Scikit-learn | OpenCV |
Tesseract

Model Deployment & Engineering Tools

FastAPI | Flask | Docker |
Git | Linux | Shell
Scripting

Cloud

AWS (S3, Lambda, EC2,
API Gateway,
DynamoDB,
CloudWatch, IAM)

Data & Databases

SQL | MongoDB

Programming Language

Python

EXPERIENCE

Software Engineer, Tech Mahindra (Client: Nestlé)

09/2024 – Present | Bengaluru, India

- Analyzed structured service telemetry datasets using **Python**, **Pandas**, and **NumPy** to identify behavioral patterns and support anomaly investigation across production environments.
- Designed and maintained monitoring dashboards across **20+ distributed services** using **Grafana** and **New Relic**, improving observability and accelerating incident detection workflows.
- Processed and validated large-scale system logs using **Linux CLI pipelines**, improving data reliability for downstream analytics and reducing issue resolution time.

Internship, Python Developer, Supreme Technologies

06/2024 – 09/2024 | Mohali, India

- Performed exploratory data analysis on a 5,000+ row employee salary dataset using **Pandas**, visualizing feature distributions and correlations using **Matplotlib** and **Seaborn** to surface key patterns.
- Handled missing values, engineered features, and built a structured preprocessing pipeline including encoding, imputation, and feature scaling for model readiness.
- Trained and evaluated regression models, improving predictive performance by upgrading from **Linear Regression** to **Random Forest**, reducing prediction error over baseline.

PERSONAL PROJECTS

AniLens: Animal Image Classification System 🐾,

02/2026 – 04/2026

TensorFlow, MobileNetV2, Flask

- Designed a binary image classifier on a 16K-image dataset, progressing from a from-scratch CNN (88% accuracy) to a MobileNetV2 transfer-learning model with staged fine-tuning, reaching 98% accuracy and 0.98 F1.
- Built an embedding-based out-of-distribution detection and multi-crop inference voting to make predictions more robust on ambiguous or unseen inputs.
- Containerized the application with Docker and deployed it on Hugging Face Spaces, with a Flask API serving real-time predictions.

HR Intelligence Platform - Attrition Prediction, Salary Insights,

03/2025 – 05/2025

Scikit-learn, Random Forest, Flask, MongoDB

- Ingested and processed 5K+ HR records from CSV and XLS sources, performing data cleaning, normalization, and structured feature engineering prior to persistence in MongoDB.
- Trained and evaluated a Decision Tree classifier for attrition prediction, achieving ~88–90% accuracy, and performed feature importance analysis to identify key attrition drivers.
- Evaluated model performance using precision, recall, and F1 metrics to validate classification reliability beyond accuracy.
- Served predictions via a Flask API for real-time attrition risk inference.

CERTIFICATIONS

- AWS Certified Machine Learning Engineer - Associate [↗](#)
- AWS Certified Solutions Architect - Associate [↗](#)
- AWS Certified AI Practitioner [↗](#)
- AWS Certified Cloud Practitioner [↗](#)
- Machine Learning (Coursera) [↗](#)

EDUCATION

Chandigarh University,
Bachelor of Engineering in Computer Science and Engineering

2020 – 2024 | Gharuan, Punjab

Extracurricular

- Co-authored and published a research paper on Card Transaction Anomaly Detection in digital payment systems, IJARESM, December 2023. Conducted literature review and comparative analysis of fraud detection methodologies for credit card transaction security.
- Participated in a college-level hackathon, collaborating in a team to design and prototype a real-time web application using Flask and SocketIO within a time-constrained environment.